Østfold University College

A Systematic Literature Review of DevOps and Continuous integration/delivery in quantum software engineering

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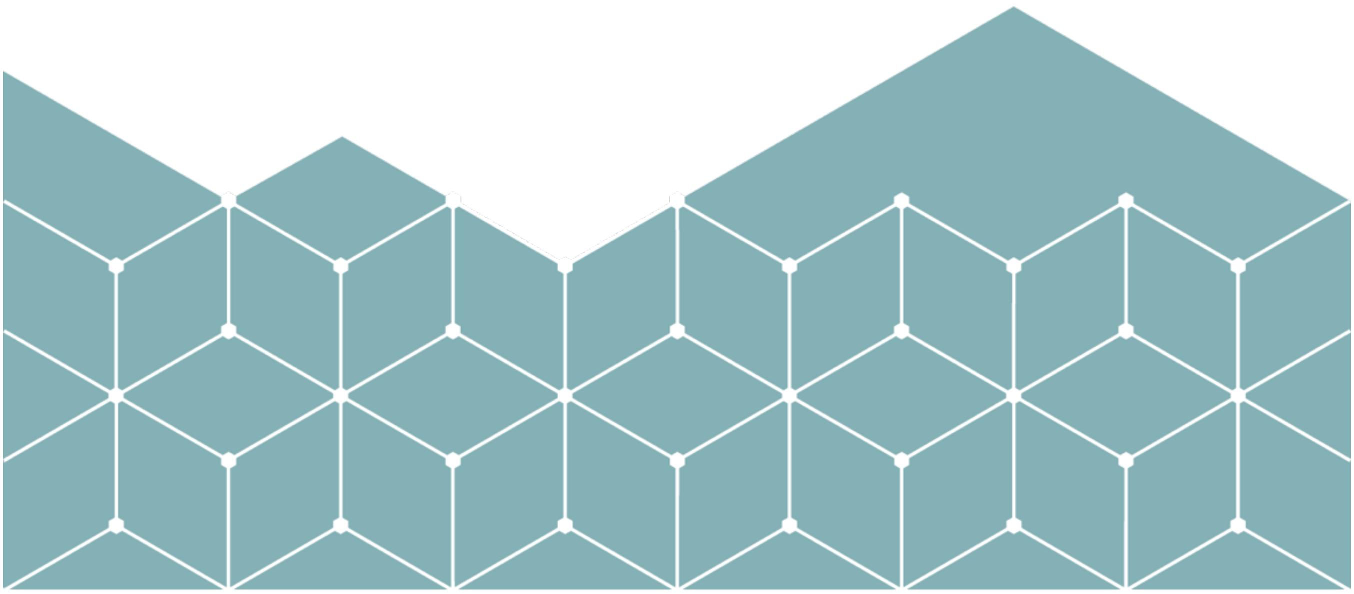
Kadir Amini, Berat Murseli

Avdeling for Informasjonsteknologi
Høgskolen i Østfold
Halden
April 30, 2026



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Abstract

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Chapter 1

Introduction

Quantum computing has developed rapidly in recent years, with cloud-accessible quantum processors now available from IBM, Google, Amazon, and Microsoft. However, the software engineering side of the field is still relatively immature. Much of the existing research focuses on algorithms, hardware, and performance, while less attention has been given to development workflows, testing pipelines, deployment practices, and automation in quantum software projects.

DevOps has quickly become a dominant part in software engineers ability to deliver software quickly and reliably. continuous integration (CI), automated testing, containerized deployment, and more are standard practices for software development. If quantum software is ever to go beyond science research and become deployable technology, the ability to deliver, maintain, and develop it will require DevOps.

quantum computers are different from classical computers at a fundamental level. Quantum computer are probabilistic due to relying on quantum mechanics to represent the binary values. Quantum computers are sensitive to noise, qubit calibration, decoherence, gate errors, SDK verion, and more. All this results in the same program potentially behaving differently on when and where it is executed.

1.1 Goal

A key objective of this SLR is to identify research gaps in the field of quantum computing, more specifically within DevOps and CI/CD. By identifying gaps in the existing research, this paper aims to provide direction for future studies and contribute to helping quantum computing researchers move closer to practical solutions in this field.

A second objective is to provide a theoretical framework for how DevOps and CI/CD practices can be integrated into the field of quantum computing. This aims to provide an understanding of how development, testing, deployment, and continuous integration processes can be adapted to make use of the strengths of quantum software development. By doing so, this paper contributes ideas on how quantum computing can reduce the gap between concept and reality.

1.2 Background

The purpose of this project is to gain a better understanding of how quantum computing works. With the advanced technology available today, we wanted to explore how modern computing technology is built, what principles lie behind the development of quantum computing, and how this emerging field may influence future technological development.

Understanding the concept of quantum computing is a large field, and it would take too much time to cover everything. Therefore, we chose to focus more on the field of DevOps and CI/CD, especially understanding the fundamentals behind the integration of quantum computing. We wanted to explore how existing applications are adapting to and using this technology today, and whether practical use cases already exist.

1.3 Research Questions

This review is guided by the following research questions:

- **RQ1:** How does quantum computing DevOps compare to standard DevOps?
- **RQ2:** What challenges exist in implementing continuous integration and continuous delivery for quantum software?
- **RQ3:** What are the main barriers preventing the adoption of standard DevOps practices in quantum software engineering?

These questions allow the review to move beyond a simple mapping of papers. They make it possible to compare quantum software workflows with classical DevOps, identify technical and organizational challenges, and explain why adoption is currently limited.

Chapter 2

Methodology

snakke om kitchanham som en introduction

2.1 Review Design

"This project follows a systematic literature review methodology consisting of planning, conducting, and reporting phases. The review process is designed to ensure transparency, reduce bias, and allow a structured synthesis of existing research."

2.2 PICOC framing

PICOC stands for Population, Intervention, Comparison, Outcome, and Context. These elements provide structure and help define what the review is about, what type of information we are trying to gather, and what we want to compare based on the research topic.

In our review, we chose to use these elements to help justify the research focus and identify the most important information based on the available papers. This helps improve the quality of the review and ensures that the selected studies are relevant for future work.

This is our PICOC table:

Element	**Definition for this review**
Population	Quantum software projects, quantum programs, developers, and teams using quantum SDKs or quantum development workflows.
Intervention	DevOps practices, CI/CD, workflow automation, testing pipelines, version control, reproducibility, and related engineering methods.
Comparison	Standard DevOps or classical software engineering practices when a comparison is possible.
Outcome	Productivity, reliability, maintainability, automation level, correctness, reproducibility, and identified barriers.
Context	Quantum software engineering in academia or industry, including simulator-based and hardware-based settings.

Table 2.1: PICOC table

2.3 Search strategy

To cover the topics of quantum software and quantum computing, we selected several academic databases for our systematic literature review. These databases were chosen because they provide access to high-quality, peer-reviewed research relevant to our study.

- **ACM**
- **IEEE Xplore**
- **Science Direct**
- **Springer Link**
- **Wiley**
- **Taylor And Francis**

We started with Google Scholar and tested different search strings. Several of the first search strings returned a very large number of papers because they were too broad. Therefore, we made the search string more specific and tested different versions until we found a more complete search string that returned around 4,000 papers. This gave us a better basis for finding relevant and high-quality papers. After that, we used the chosen search string to search in the different databases we had selected.

Our selected search string is: **("quantum software" OR "quantum program" OR "quantum programming" OR "quantum computing") AND ("DevOps" OR "CI/CD")**

2.4 Inclusion and exclusion criteria

As part of our review method, we needed to define criteria for what should be included and excluded in our SLR. Since our language knowledge is mainly based on English, we chose to include only papers written in English and exclude papers written in other languages. This makes it possible for us to read the papers and understand the information.

We also chose to include only peer-reviewed papers, as this helps ensure a higher level of quality. Papers that did not meet the required quality level were excluded.

There is a large amount of research on quantum computing, and some papers focus mainly on quantum physics. These papers were excluded because our review focuses specifically on quantum software development and software engineering.

We wanted to focus on newer research in the field, so we excluded papers published before 2020. This helps ensure that the review includes more recent developments and innovations.

To make our research more relevant to the search field, we also excluded papers that did not focus on topics such as DevOps, CI/CD, workflow automation, and testing pipelines.

This helped us narrow the review and identify papers that were more closely related to our research topic.

Inclusion criteria	**Exclusion criteria**
Written in English	Non-English publications
Peer-reviewed articles or clearly relevant scholarly papers	Non-peer-reviewed material with no academic value
Focuses on quantum software development or quantum software engineering	Pure quantum physics or hardware papers with no software relevance
Published between 2020 and 2026	Published before 2020 unless needed for background only
Discusses DevOps, CI/CD, workflow automation, testing pipelines, or closely related practices in a quantum context	No meaningful connection to DevOps or software-engineering practice in a quantum context

Table 2.2: Inclusion and Exclusion Criteria Table

2.5 Study selection process

1. Stage 1: Title and abstract screening to remove clearly irrelevant studies. 2. Stage 2: Full-text screening using the inclusion and exclusion criteria. 3. Stage 3: Final set of papers prepared for data extraction and synthesis. 4. At each stage, exclusion reasons should be documented to improve transparency.

2.6 Quality assessment

2.7 Data extraction plan

2.8 Data synthesis plan

Chapter 3

Results

3.1 Overview of Included Studies

3.2 RQ1: DevOps Practices in Quantum Software Enginee

3.3 RQ2: CI/CD and Automation Challenges

3.4 RQ3: Barriers to Adoption

Chapter 4

Discussion

4.1 Summary

Chapter 5

Conclusion

Bibliography